Software Requirements Specification (SRS)

Project: Event Management System
Version: 1.0
Team: Team 7

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Revision history

Version	Date	Author	Change summary	Approval
1.0	03-09-2026	Team 7	Initial SRS for Event Management System based on the supplied SRS template.	

Approvals

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the Event Management System. It serves as a common reference for the student development team, reviewers, testers, and evaluators. The document describes what the system shall do, the interfaces it shall provide, quality expectations, security controls, and the verification criteria used to judge completion.

1.2 Scope

The system covers creation and management of events, publication of event information, participant registration, participant record management, event scheduling, registration monitoring, and basic reporting. The scope is limited to the software system and its application interfaces; physical venue management, external payment settlement, and institution-wide identity infrastructure are outside the defined baseline unless added through a later approved change.

1.3 Audience

Primary readers are the project team, course instructor/evaluator, testers, and future maintainers of the software.

1.4 Definitions

- EMS: Event Management System.
- Organizer: Authorized user who creates and manages events.
- Participant: User who views events and registers for them.
- Administrator: Authorized user responsible for system-level administration and oversight.
- Event: A scheduled activity represented in the system with details such as title, date, time, venue, and capacity.
- Registration: A participant record indicating intent to attend a specific event.
- API: Application Programming Interface used for communication between software components.
- RTM: Requirements Traceability Matrix connecting requirements to design modules and test cases.

2. Overall Description

2.1 Product Perspective

The Event Management System is a web-based application. The user interface is implemented using React.js. Server-side application services use Node.js, while Python/Django is used for the defined Python-based administrative or data-management components. The system is intended to centralize event information and registrations so that organizers can manage events and participants can discover and register for them through a consistent interface.

2.2 Major Product Functions

- User account authentication and role-based access.
- Event creation, editing, publishing, and cancellation.
- Event listing, search, filtering, and detailed viewing.
- Participant registration and registration status handling.
- Capacity control and prevention of invalid registrations.
- Organizer access to participant lists and event-level registration information.
- Administrator user and event oversight.
- Event updates and status changes visible to affected users.

• Basic event and registration reports.
• Validation, audit logging, and secure handling of user information.

2.3 User Roles and Characteristics
• Participant: General user; browses events, views event details, registers, and reviews personal registrations.
• Organizer: Creates and manages events, reviews registrations, updates event details, and manages participant information for owned events.
• Administrator: Manages users and oversees event records, system usage, and administrative reporting.

2.4 Operating Environment
The application is designed to operate through a modern web browser on desktop or mobile devices. The software stack consists of React.js for the frontend, Node.js for application/API services, and Python/Django for defined Python-based modules, with a persistent database accessed through the server-side components.

2.5 Constraints
• The project shall use the team-approved stack: JavaScript with React.js and Node.js, plus Python with Django.
• The system shall enforce role-based access so organizer and administrator functions are not exposed to ordinary participants.
• The baseline project shall not assume external payment, SMS, or third-party identity services unless separately integrated and documented.
• The implementation shall provide a responsive web interface suitable for common desktop and mobile browser sizes.

3. External Interface Requirements

3.1 User Interfaces

The UI shall provide clear navigation for event discovery, event management, registration, and administration. Forms shall display field labels, validation messages, success/error feedback, and confirmation prompts for destructive actions such as cancellation or deletion. The interface should remain understandable at common desktop and mobile viewport sizes.

3.2 Hardware Interfaces

No dedicated hardware interface is required for the baseline system. The application is accessed using a standard device with a display, keyboard/touch input, and network connectivity.

3.3 Software Interfaces

- React.js frontend communicates with server-side APIs for authentication, events, registrations, and reports.
- Node.js application services expose the required application endpoints and business logic.
- Django provides the specified Python-based administrative/data-management functionality.
- The system communicates with its database through the server-side application layer.

3.4 Communications

All client-server communication in a deployed environment shall use HTTPS. API requests shall carry authenticated session or token information as defined by the implementation. The system shall handle invalid requests and server errors using appropriate HTTP responses and user-facing messages.

4. System Features

4.1 Functional Requirements

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case	Comments / Dependencies
EMS-F-001	The system shall allow a user to register an account using the required registration fields.	Functional	High	Participant / Organizer / Administrator / System	Account form accepts valid data and rejects invalid/missing required values. Test: TC-REG-01	See RTM / design module
EMS-F-002	The system shall authenticate users and identify their assigned role after successful login.	Functional	High	Participant / Organizer / Administrator / System	Valid credentials create an authenticated session with the correct role. Test: TC-AUTH-01	See RTM / design module
EMS-F-003	The system shall allow an organizer to create an event with title, description, date, time, venue, and capacity.	Functional	High	Participant / Organizer / Administrator / System	A valid event is saved and appears in the organizer event list. Test: TC-EVT-01	See RTM / design module
EMS-F-004	The system shall allow an organizer to edit the details of an event that they are authorized to manage.	Functional	High	Participant / Organizer / Administrator / System	Updated event information is persisted and displayed to users. Test: TC-EVT-02	See RTM / design module
EMS-F-005	The system shall allow an organizer or administrator to publish or cancel an event according to their permissions.	Functional	High	Participant / Organizer / Administrator / System	Status changes are persisted and reflected in event listings. Test: TC-EVT-03	See RTM / design module

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case	Comments / Dependencies
EMS-F-006	The system shall display a list of available events with basic summary information.	Functional	High	Participant / Organizer / Administrator / System	Users can view event titles, dates, venues, and current status. Test: TC-VIEW-01	See RTM / design module
EMS-F-007	The system shall allow participants to search or filter events using supported event information such as title, date, or category.	Functional	Medium	Participant / Organizer / Administrator / System	Entered search/filter criteria produce matching results or a clear no-results state. Test: TC-SEARCH-01	See RTM / design module
EMS-F-008	The system shall display detailed information for a selected event, including registration availability.	Functional	High	Participant / Organizer / Administrator / System	Event page shows required details and whether registration is open/full/closed. Test: TC-VIEW-02	See RTM / design module
EMS-F-009	The system shall allow an authenticated participant to register for an event that is open and not full.	Functional	High	Participant / Organizer / Administrator / System	Successful registration creates one participant-event registration record. Test: TC-REG-02	See RTM / design module
EMS-F-010	The system shall prevent duplicate registration by the same participant for the same event.	Functional	High	Participant / Organizer / Administrator / System	A second registration attempt is rejected without creating a duplicate record. Test: TC-REG-03	See RTM / design module
EMS-F-011	The system shall prevent registration when an event has reached its configured capacity.	Functional	High	Participant / Organizer / Administrator / System	Registration is blocked when available capacity is zero. Test: TC-REG-04	See RTM / design module
EMS-F-012	The system shall allow participants to view their active and past registrations.	Functional	Medium	Participant / Organizer / Administrator / System	The user can view a list of their registration records with event status. Test: TC-REG-05	See RTM / design module
EMS-F-013	The system shall allow authorized organizers to view and manage the participant list for events they own.	Functional	High	Participant / Organizer / Administrator / System	Organizer can retrieve participant records only for authorized events. Test: TC-ORG-01	See RTM / design module
EMS-F-014	The system shall allow an administrator to view and manage user and event records according to administrative permissions.	Functional	High	Participant / Organizer / Administrator / System	Administrator actions are available only to authorized administrators. Test: TC-ADM-01	See RTM / design module
EMS-F-015	The system shall provide basic reports showing event and registration information for authorized users.	Functional	Medium	Participant / Organizer / Administrator / System	Report generation returns correct counts/details for the selected scope. Test: TC-REP-01	See RTM / design module
EMS-F-016	The system shall record important event and registration status changes in an audit log.	Functional	Medium	Participant / Organizer / Administrator / System	Defined actions generate timestamped audit entries associated with the acting user. Test: TC-AUD-01	See RTM / design module
EMS-F-017	The system shall validate date/time and capacity inputs before saving an event.	Functional	High	Participant / Organizer / Administrator / System	Past/invalid schedule values and invalid capacities are rejected with validation feedback. Test: TC-VAL-01	See RTM / design module

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case	Comments / Dependencies
EMS-F-018	The system shall display clear error messages when an operation fails because of invalid input, insufficient permission, or a server-side error.	Functional	Medium	Participant / Organizer / Administrator / System	Failure states provide an understandable message and do not create partial/duplicate records. Test: TC-ERR-01	See RTM / design module

5. Non-Functional Requirements

The following measurable quality requirements use the EMS-NF-### convention.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
EMS-NF-001	Typical user-facing API operations shall respond within 3 seconds for at least 90% of requests under the project test load.	Performance	High	90th percentile response time <= 3s in production-like test. Test: TC-PERF-01
EMS-NF-002	The system shall achieve at least 99.0% monthly availability excluding planned maintenance for the project deployment environment.	Reliability	High	Availability report meets or exceeds 99.0%. Test: TC-OPS-01
EMS-NF-003	The application shall enforce role-based authorization on protected operations and shall not rely only on client-side hiding of controls.	Security	High	Unauthorized API requests are rejected. Test: TC-SEC-01
EMS-NF-004	The system shall maintain timestamped application and security logs for important authentication, event, registration, and administrative actions.	Auditability	Medium	Required events appear in logs with timestamp and actor information. Test: TC-LOG-01
EMS-NF-005	The user interface shall support responsive layouts and basic accessibility practices, including meaningful labels, readable feedback, and keyboard-accessible controls where applicable.	Usability/Accessibility	Medium	UI review checklist passes on supported screen sizes and keyboard interactions. Test: TC-UX-01

5.1 Security

5.1.1 Security Objectives

1. Protect user and event-management data from unauthorized access or modification by enforcing authenticated, role-based access to protected functions.
2. Maintain integrity and accountability of event and registration operations by validating requests, recording security-relevant actions, and protecting client-server communication.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case
EMS-SR-001	The system shall require authenticated access for participant, organizer, and administrator actions that read or modify protected records.	Security	High	Unauthenticated protected requests are rejected. Test: TC-SEC-01
EMS-SR-002	The system shall enforce server-side role and ownership checks for organizer and administrator operations.	Security	High	A user cannot access or modify records outside their authorization scope. Test: TC-SEC-02

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case
EMS-SR-003	The system shall use HTTPS/TLS for deployed client-server communication.	Security	High	Network inspection confirms encrypted transport in deployment. Test: TC-SEC-03
EMS-SR-004	The system shall validate and sanitize user-provided input before processing or storing it.	Security	High	Invalid/malicious test inputs are rejected or safely handled without unintended execution. Test: TC-SEC-04
EMS-SR-005	The system shall not store passwords in plaintext and shall protect credentials using an appropriate password-hashing mechanism provided by the application framework.	Security	High	Credential storage inspection confirms password hashes rather than plaintext passwords. Test: TC-SEC-05

6. Quality Attributes & Acceptance Tests

The project shall be considered ready for acceptance when all high-priority functional and security requirements are implemented and their referenced test cases pass, no critical defects remain in the core workflows, and the RTM shows traceability from requirements to tests.

Quality Attribute	Acceptance Approach
Functional correctness	Execute registration, event creation/editing, event discovery, registration, organizer management, administration, and reporting test suites.
Performance	Run a representative test load and verify EMS-NF-001 response target.
Reliability	Verify service startup, recovery from common failures, and availability target for the project deployment environment.
Security	Execute authentication, authorization, input-validation, transport-security, and credential-storage tests.
Usability/Accessibility	Perform UI review on supported screen sizes, navigation, labels, validation feedback, and keyboard interaction.
Auditability	Trigger defined auditable actions and verify timestamped records are produced.

7. System Models and Diagrams

7.1 UML Use-Case Diagram 1 - Core Event and Registration Flow

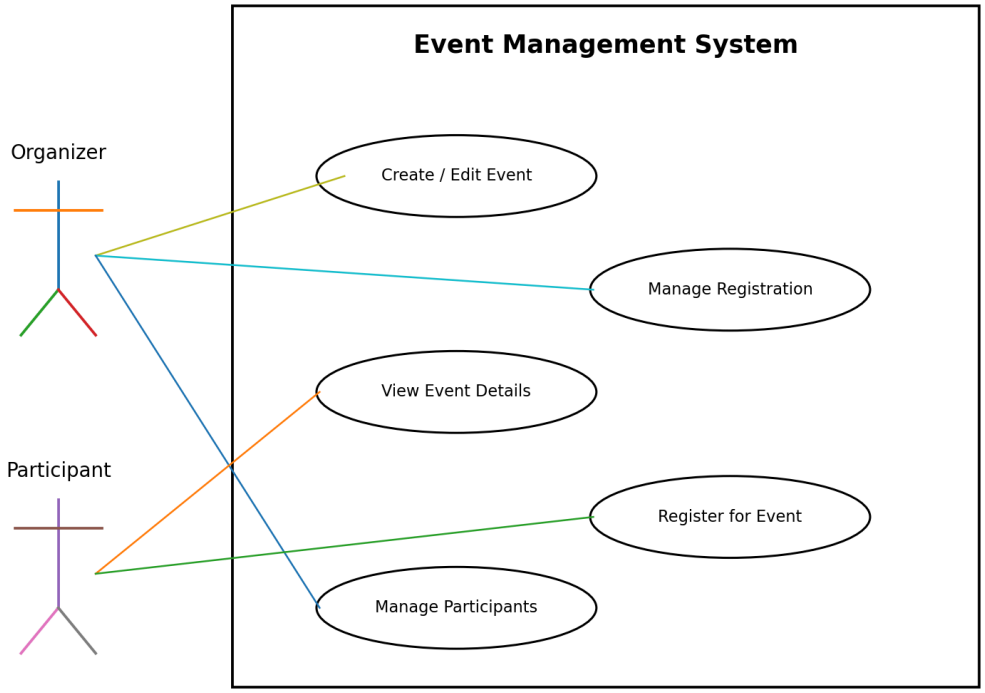


Figure 1. Core use cases involving Organizer and Participant.

7.2 UML Use-Case Diagram 2 - Administration and Oversight

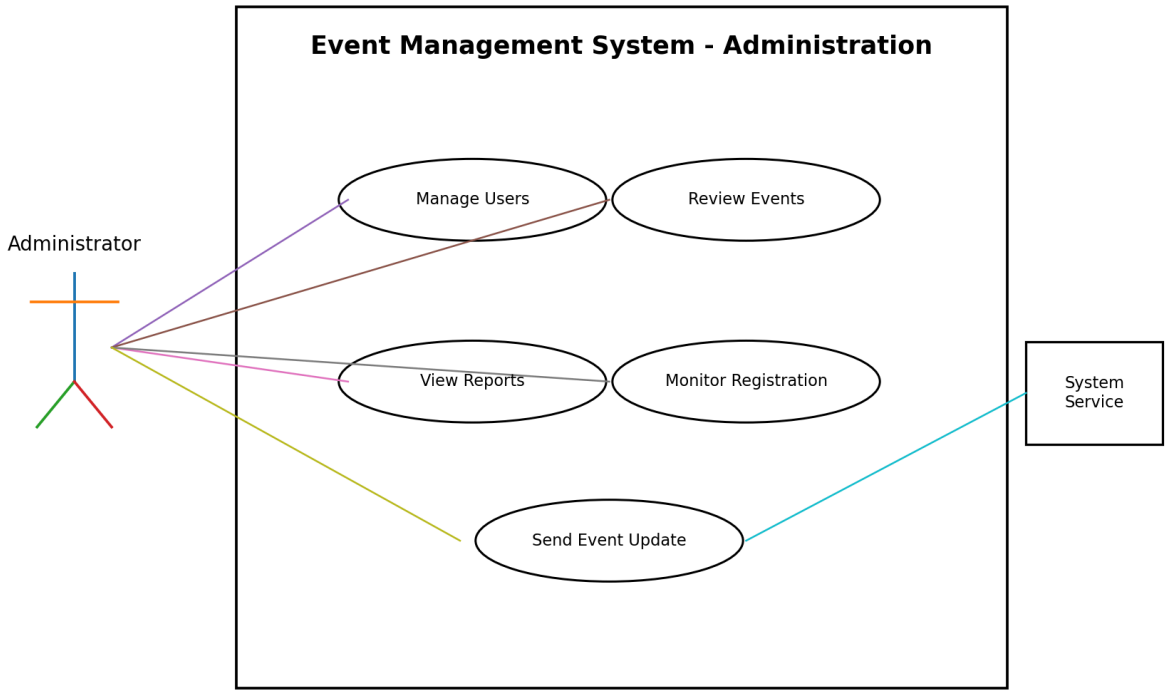


Figure 2. Administrative and system oversight use cases.

7.3 Primary Workflow Summary

1. Participant creates an account or signs in.
2. Participant browses or searches available events and opens an event detail page.
3. Participant submits registration when the event is open and capacity is available.
4. System validates the participant and event state, prevents duplicates, and stores the registration.
5. Organizer views and manages the participant list for the event and can update event information.
6. Administrator oversees user/event records and generates basic reports.

8. Requirements Traceability Matrix (RTM)

The matrix below provides baseline traceability. Design-module names and test IDs are project-facing labels and may be refined during implementation without changing the requirement intent.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
EMS-F-001	User registration	4 / DS-Auth-01	Auth & User Module	TC-REG-01	N	
EMS-F-002	User authentication	4 / DS-Auth-02	Auth & User Module	TC-AUTH-01	N	
EMS-F-003	Create event	4 / DS-Event-01	Event Module	TC-EVT-01	N	
EMS-F-004	Edit event	4 / DS-Event-02	Event Module	TC-EVT-02	N	
EMS-F-005	Publish/cancel event	4 / DS-Event-03	Event Module	TC-EVT-03	N	
EMS-F-006	Event listing	4 / DS-Event-04	Event Discovery UI	TC-VIEW-01	N	
EMS-F-007	Search/filter events	4 / DS-Event-05	Event Discovery UI	TC-SEARCH-01	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
EMS-F-008	Event details / availability	4 / DS-Event-06	Event Discovery UI	TC-VIEW-02	N	
EMS-F-009	Participant registration	4 / DS-Reg-01	Registration Module	TC-REG-02	N	
EMS-F-010	Prevent duplicate registration	4 / DS-Reg-02	Registration Module	TC-REG-03	N	
EMS-F-011	Capacity control	4 / DS-Reg-03	Registration Module	TC-REG-04	N	
EMS-F-012	View personal registrations	4 / DS-Reg-04	Participant Module	TC-REG-05	N	
EMS-F-013	Organizer participant management	4 / DS-Org-01	Organizer Module	TC-ORG-01	N	
EMS-F-014	Administrator management	4 / DS-Admin-01	Administration Module	TC-ADM-01	N	
EMS-F-015	Basic reports	4 / DS-Report-01	Reporting Module	TC-REP-01	N	
EMS-F-016	Audit logging	4 / DS-Audit-01	Audit/Logging Module	TC-AUD-01	N	
EMS-F-017	Input validation	4 / DS-Validation-01	Validation Module	TC-VAL-01	N	
EMS-F-018	Error handling	4 / DS-Error-01	API/UI Error Handler	TC-ERR-01	N	
EMS-NF-001	API response target	5 / DS-Perf-01	API Services	TC-PERF-01	N	
EMS-NF-002	Availability target	5 / DS-Reliability-01	Deployment / Services	TC-OPS-01	N	
EMS-NF-003	Authorization enforcement	5 / DS-Sec-01	Auth Middleware / API	TC-SEC-01	N	
EMS-NF-004	Timestamped audit/security logs	5 / DS-Log-01	Audit/Logging Module	TC-LOG-01	N	
EMS-NF-005	Responsive/accessibility	5 / DS-UX-01	React UI	TC-UX-01	N	
EMS-SR-001	Authenticated protected access	5.1 / DS-Sec-01	Auth Middleware / API	TC-SEC-01	N	
EMS-SR-002	Role and ownership checks	5.1 / DS-Sec-02	Auth Middleware / API	TC-SEC-02	N	
EMS-SR-003	HTTPS/TLS	5.1 / DS-Sec-03	Deployment / API	TC-SEC-03	N	
EMS-SR-004	Input validation/sanitization	5.1 / DS-Sec-04	Validation Module	TC-SEC-04	N	
EMS-SR-005	Password hashing	5.1 / DS-Sec-05	Auth & User Module	TC-SEC-05	N	

Appendix A - Revision History & Approval

Version	Date	Author	Change Summary
1.0	03-09-2026	Team 7	Initial SRS for Event Management System based on the supplied SRS template.
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Course Coordinator			
Project Team Member	Ashmita Chaki	Ashmita Chaki chakiashmita@gmail.com	03-09-2026
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